R13

Code No: 117BX

JAWAHARLAL NEHRU TECHNOLOGICAL UNIVERSITY HYDERABAD
B. Tech IV Year I Semester Examinations, July/August - 2023
COMPUTER GRAPHICS
(Computer Science and Engineering)

Time: 3 Hours **Max. Marks: 75**

Note: i) Question paper consists of Part A, Part B.
ii) Part A is compulsory, which carries 25 marks. In Part A, Answer all questions.
iii) In Part B, Answer any one question from each unit. Each question carries 10 marks and may have a, b as sub questions.

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PART – A

(25 Marks)

1.a)	Where is flood algorithm used?	[2]
b)	Define work station.	[3]
c)	List out clipping algorithms.	[2]
d)	What are Shear Transformations?	[3]
e)	What are the advantages of Beizer curves?	[2]
f)	Define 3D viewing.	[3]
g)	What is depth sorting?	[2]
h)	Why do we need polygon rendering methods?	[3]
i)	List out computer animation languages.	[2]
j)	Write about key frame systems.	[3]

PART – B

(50 Marks)

2.a) List the operating characteristics of
i) Raster refresh systems (ii) Vector refresh systems
iii) Plasma panel (iv) LCDs.

b) How much time is spent in scanning across each row of pixels during screen refresh on a raster system with a resolution of 1280 by 1024 and a refresh rate of 60 frames per second? Assume horizontal and vertical retrace times are negligible? [5+5]

OR

3.a) Explain the mathematical procedure in deciding the points on the periphery of the ellipse using mid-point ellipse algorithm?

b) Illustrate the properties of ellipse which are considered in efficient generation of ellipse using mid-point method. [5+5]

4.a) Give the transformation matrix to rotate a point about an arbitrary point.

b) Give the transformation matrix for the following
i) To shift left by 2 units and due to rotate by 45^0 clockwise.
ii) To reflect with respect to y =-x axis. [5+5]

OR

5.a) What are the advantages and disadvantages of Sutherland - Hodgeman algorithm?

b) How the 4-bit outcodes are computed which are used is Cohen-Sutherland algorithm?[5+5]

6. Derive the transformation matrix for B-spline curve. [10]

OR

7.a) Derive the transformation matrix for parallel projections.

b) Give the transformation matrix in 3D for the following.

i) To rotate a point about the y-axis by 45^0 in counter clock wise direction.

ii) Reflect a point about x-axis. [5+5]

8.a) Explain depth-buffer algorithm for hidden surface removal.

b) Write an algorithm to display the visible surfaces of a convex polyhedron using the depth-sort (painter's) algorithm? [5+5]

OR

9. Describe in brief about area sub division algorithm. [10]

10.a) List the typical tasks for which the animation functions are defined in animation languages.

b) What are the steps in the design of animation sequence? [5+5]

OR

11. Describe in brief about Raster animation. [10]

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